

Name: _____

These questions assess your understanding of the material from Chapters 3, 4 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

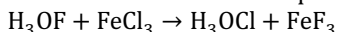
1. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Hérout process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

2. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 381. kg of compound A and 172. kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 79%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

3. A way to produce solid iron (III) fluoride is by combining aqueous hydronium fluoride with aqueous iron (III) chloride. The chemical reaction is represented by the following unbalanced chemical equation:



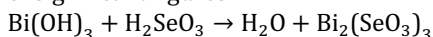
Sophia wants to produce 29.1 mg of iron (III) fluoride. She pours 12.5 mL of a 0.2664-M solution of aqueous hydronium fluoride into a beaker. What volume of a 0.219-M solution of aqueous iron (III) chloride is needed to react with the hydronium fluoride solution to produce 29.1 mg of iron (III) fluoride? If you find that 12.5 mL of the hydronium fluoride solution is insufficient, report the required volumes of both reactants required to produce the 29.1 mg of iron (III) fluoride.

ANSWER: _____

4. Justin conducted some measurements and determined that 0.733 moles of nitrogen gas participated in a chemical reaction. Calculate the mass of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

5. Given this unbalanced chemical equation, calculate and report the percent yield of $\text{Bi}_2(\text{SeO}_3)_3$ if 5.805 g of $\text{Bi}_2(\text{SeO}_3)_3$ is produced from the reaction of 10.2 g of $\text{Bi}(\text{OH})_3$ and 9.922 g of H_2SeO_3 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

6. Robert conducted some measurements and determined that 603. g of oxygen gas participated in a chemical reaction. Calculate the number of moles of oxygen gas that participated in the chemical reaction.

ANSWER: _____

7. A particular moon has a mass of 2.0×10^{22} kg and a particular planet has a mass of 3.7×10^{26} g. What is the mass of 1 mole of this particular moon? Ensure your answer has the correct number of significant figures.

ANSWER: _____

8. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

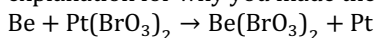
ANSWER: _____

9. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 40.9 g.

Compound	Initial Mass	Final Mass
A	11.5 g	6.1 g
B	47.7 g	0 g

ANSWER: _____

10. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

11. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide and oxygen reacts with water to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

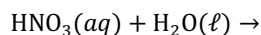
12. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

13. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

14. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Be sure your equation is balanced and contains state symbols after every reactant and product.



ANSWER: _____

15. What is the oxidation number of chlorine in antimony trichloride?

ANSWER: _____

16. Linda needs to make 128.28 mL of aqueous lithium sulfide (Li_2S) working solution. She plans to do this by adding distilled water to a 83.36-mL stock solution of 0.64-M aqueous lithium sulfide. What will be the concentration of her working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

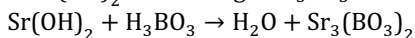
17. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

18. An oil tanker has a mass of 4.687×10^8 kg and a cooking pot has a mass of 2.06×10^{-1} kg. What is the mass of 1 mole of cooking pots? Ensure your answer has the correct number of significant figures.

ANSWER: _____

19. Given this unbalanced chemical equation, calculate and report the theoretical yield of H_2O formed by the mixing of 26.8 g of $\text{Sr}(\text{OH})_2$ with 14.1 g of H_3BO_3 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

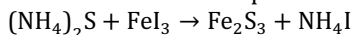
20. John has a 0.212-M solution of magnesium sulfate (MgSO_4). If he needs to add 0.547 g of magnesium sulfate to a reaction flask, how much of the solution should he pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

21. Justin is instructed to dissolve 4.002 g of iron (III) chloride in 128. mL of a 887.-mM aqueous solution of ammonium phosphate. Calculate the molarity of the chloride anion in the solution after the reaction has occurred. Assume the volume of the solution does not change when chloride is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

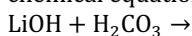
22. A way to produce solid iron (III) sulfide is by combining aqueous ammonium sulfide with aqueous iron (III) iodide. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Joseph pours 16.38 mL of a 0.064-M solution of aqueous ammonium sulfide and then adds 12.6 mL of a 0.2801-M solution of aqueous iron (III) iodide. He then stirs the mixture and notices iron (III) sulfide precipitating out of the solution. What is the mass of iron (III) sulfide that he should expect the reaction to yield?

ANSWER: _____

23. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

24. As part of his routine quality-control checks, Thomas, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hydrobromic acid solution. He takes a 50.0-mL sample of hydrobromic acid solution and titrates it with an aluminum hydroxide solution, which he made by dissolving 1.94 g of solid aluminum hydroxide in 1.11 L of distilled water. If Thomas finds that it takes 105.6 mL of the aluminum hydroxide solution for the titration to reach the equivalence point, what molarity of the hydrobromic acid solution does Thomas calculate?

ANSWER: _____

25. A strip of solid nickel metal is put into a beaker of 0.161-M AuBrO solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

26. Jeremy is performing a combustion analysis on a compound with a molecular molar mass of 100.16 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 4.78 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
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Carbon Dioxide	12.6 g
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Water	5.16 g
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Use this information to find the molecular formula of the compound.

ANSWER: _____

27. Barium hydroxide reacts with sulfurous acid to form water and barium sulfite. Calculate the mass of sulfurous acid needed to produce 13.9 g of water. Ensure your answer has the correct number of significant figures.

ANSWER: _____

28. Tyler has a 0.6312-M solution of copper (II) bromide (CuBr_2). If he pours 4.35 mL of copper (II) bromide into a reaction flask, what is the mass of copper (II) bromide that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

29. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

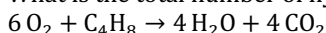
30. What major chemical species are present in an aqueous solution of 0.15-M hydronium bromide?

ANSWER: _____

31. Calculate the mass of a sample of magnesium chloride (MgCl_2) that contains 5.340×10^{18} magnesium atoms.

ANSWER: _____

32. What is the total number of hydrogen atoms on the left-hand side of this chemical equation?



ANSWER: _____

33. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 1.60 moles of A and 3.21 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

34. William prepares a solution of hydronium chloride (H_3OCl) by measuring out 34.32 mmol of hydronium chloride into a 50.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

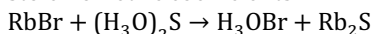
35. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

36. Anna prepares a solution of sodium phosphate (Na_3PO_4) by measuring out 3.016 g of sodium phosphate into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

37. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.



ANSWER: _____

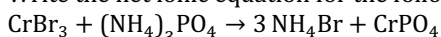
38. The chemical formula for aluminum oxide is Al_2O_3 . Calculate the molar mass of aluminum oxide. Round your answer to 5 significant figures.

ANSWER: _____

39. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

40. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:



ANSWER: _____

41. Rubidium sulfide reacts with ammonium iodide to form ammonium sulfide and rubidium iodide. If 2.50 g of ammonium sulfide is produced from the reaction of 19.2 g of rubidium sulfide and 40.29 g of ammonium iodide, calculate the percent yield of ammonium sulfide. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

42. Calculate the number of sulfur atoms in a 0.1294-g sample of magnesium sulfide (MgS).
ANSWER: _____
43. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.
 $\text{Li} + \text{CO}_2 \rightarrow \text{Li}_2\text{O} + \text{C}$
ANSWER: _____
44. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{K}(s) + \text{H}_2\text{O}(l) \rightarrow$
ANSWER: _____
45. Ammonium oxalate reacts with potassium sulfide to form ammonium sulfide and potassium oxalate. Calculate the moles of potassium sulfide needed to produce 0.05518 moles of potassium oxalate. Ensure your answer has the correct number of significant figures.
ANSWER: _____
46. Jennifer must dilute 10.7 mL of a 5.203-mM solution of aqueous hydronium sulfide ($(\text{H}_3\text{O})_2\text{S}$) until the concentration falls to 3.804 mM. She will do this by adding distilled water to the solution. How much distilled water must she add? Be sure your answer has the correct number of significant figures.
ANSWER: _____
47. Given this unbalanced chemical equation, report the maximum mass of $\text{Ni}_3(\text{BO}_3)_2$ that could be produced by the mixing of 10.48 g of $\text{Ni}(\text{OH})_2$ with 5.629 g of H_3BO_3 . Ensure that your answer has the correct number of significant figures.
 $\text{Ni}(\text{OH})_2 + \text{H}_3\text{BO}_3 \rightarrow \text{Ni}_3(\text{BO}_3)_2 + \text{H}_2\text{O}$
ANSWER: _____
48. Kyle conducted some measurements and determined that 54.4 g of radium participated in a chemical reaction. Calculate the number of moles of radium that participated in the chemical reaction.
ANSWER: _____
49. Copper (II) hydroxide reacts with hydrosulfuric acid to form water and copper (II) sulfide. Suppose 20.2 g of copper (II) hydroxide is mixed with 9.57 g of hydrosulfuric acid. Calculate and report the theoretical yield of copper (II) sulfide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.
ANSWER: _____
50. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.
ANSWER: _____